

ML Assignment 1: Polynomial Regression

Roll number: IMT2024069

Objective

Two independent polynomial-regression models were trained to predict the continuous target y . The supplied personalized training data were used for fitting and the supplied test data were used only for inference. No non-polynomial model or target transformation was used.

Method

For each dataset, the selected columns were expanded into all polynomial and interaction terms whose total degree is at most the stated degree. An ordinary least-squares linear regression was then fitted on those expanded terms. This is polynomial regression: the model remains linear in its learned coefficients while allowing nonlinear relationships in the original inputs.

Problem	Features retained	Degree	Expanded terms
var1 - turbine optimization	x_1, x_2, x_3	3	19
var2 - reservoir mapping	x_1	4	4

Model selection and validation

The assignment brief identifies the effective configurations for this personalized data: degree 3 using the first three features for var1, and degree 4 using only the first feature for var2. I used these configurations directly. As a diagnostic, 5-fold shuffled cross-validation (random seed 2024069) was performed on the training set. The scores below are validation diagnostics, not estimates on the hidden test labels. The low validation R^2 values indicate limited predictive fit; these results do not establish that the suggested configurations are empirically optimal.

Dataset	CV MSE (mean +/- SD)	CV R^2 (mean +/- SD)	Training R^2
var1	8.737 +/- 0.859	0.155 +/- 0.033	0.211
var2	37.945 +/- 7.057	0.067 +/- 0.048	0.086

Inference and submission files

After validation, each model was retrained on all 1,000 rows of its corresponding training dataset. Predictions were generated for all 1,000 rows of the matching test dataset. Each prediction file preserves the test input columns and appends the predicted y column, so row order and test-point identity are retained.

Reproducibility

The accompanying `train_and_predict.py` script uses pandas and scikit-learn. It explicitly records the roll number, chosen feature subsets, and polynomial degrees, then regenerates both CSV files from the original four supplied files. The script also reproduces the validation scores in `validation_results.csv`. The README contains the exact command to run it.